

Large conforming-wall annular benchmark

1. Problem

Fully developed incompressible axial annular flow, $R_i=0.05$ m, $R_o=0.1$ m, $\mu=0.001$ Pa s, $\rho=1$, imposed $-dp/dz=0.01$ Pa/m. No-slip circular walls.

Exact 3D reduction: $-\text{div}(\mu \text{grad } w)=G$ on a 2D cross-section.

No general 3D flow or numerical pressure solution is asserted.

2. Numerical method

Registered conforming polar Mesh2D; shared periodic angular faces.

Actual production fitted diffusion/gradient/matvec kernel.

Full 16,384 / 65,536 / 262,144 unknown CSR matrices, CG/Jacobi, zero initial state.

No ring averaging, analytic initialization or numerical force correction.

Independent NumPy face geometry/coefficient/flux, flow and wall-force audit.

Separate geometry-only staircase comparison does not certify LBM flow.

3. Results

256x64 (16,384 cells):

velocity L2 0.042030%, flow 0.026437%, pass=True

512x128 (65,536 cells):

velocity L2 0.010510%, flow 0.006612%, pass=False

1024x256 (262,144 cells):

velocity L2 0.002628%, flow 0.001653%, pass=False

Observed velocity orders: 1.999684, 1.999921

4. Wall-law scope and next steps

Spalding dimensional traction module independently verified; not yet

coupled to wall-resolved SA. Constitutive inversion is not turbulence

physical validation. Next: matching public flat-plate data and RANS wall coupling.

Raw states, references and source SHA included; research manuscript.

Quantitative full-matrix verification

Grid	Unknowns	Velocity L2 %	Flow error %	Inner-wall force %	Passed
256x64	16,384	0.042030	0.026437	0.017925	True
512x128	65,536	0.010510	0.006612	0.004482	False
1024x256	262,144	0.002628	0.001653	0.001120	False

All physical gates are strict $< 3\%$. Each wall force is checked independently.

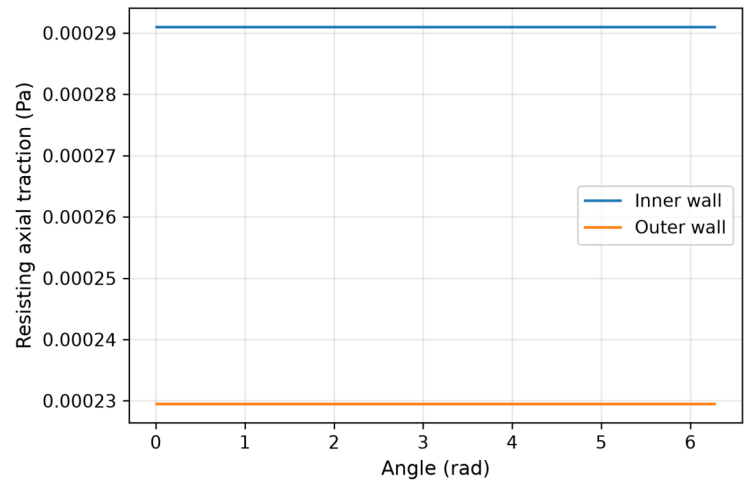
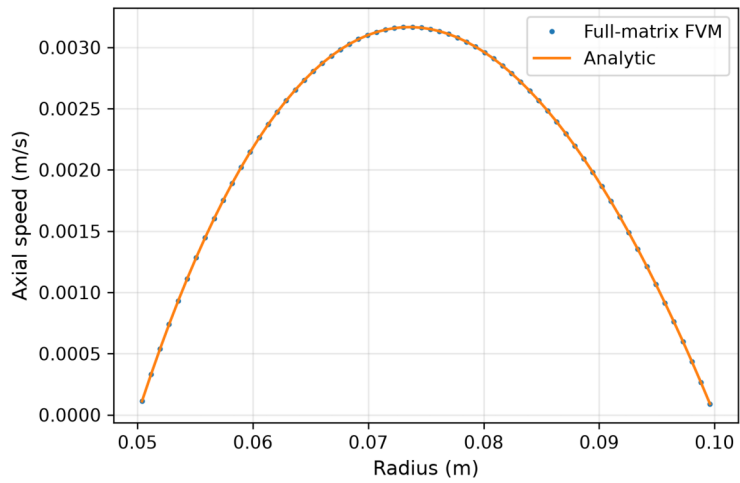
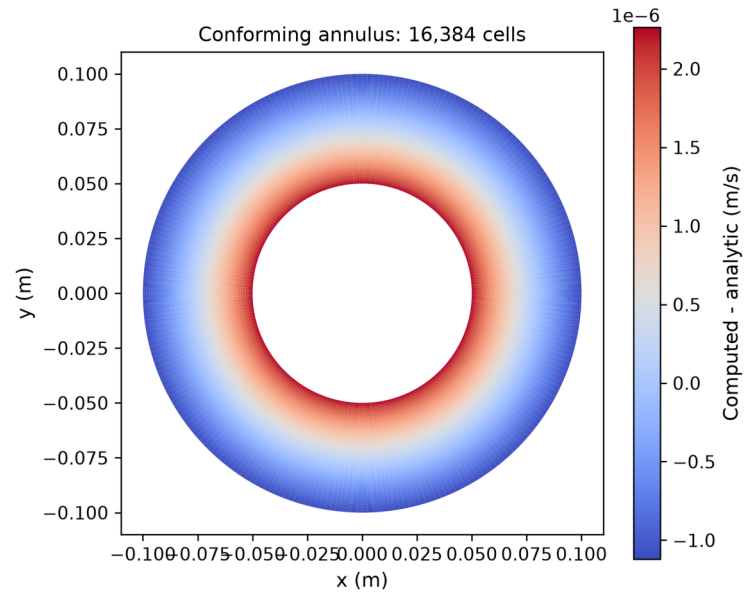
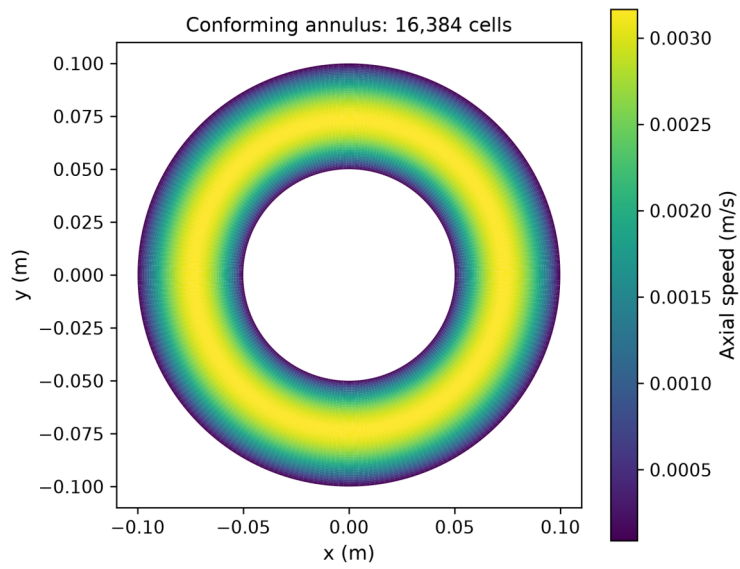
Pressure is prescribed driving data, not a computed pressure field.

Velocity is sampled at polygon-cell centers and volume-weighted.

Linear residual and global force balance are separate numerical gates.

Spalding law is a constitutive module, not a complete wall-function RANS run.

256x64 full conforming cross-section



1024x256 full conforming cross-section

